Jorge Peguero - W214777728

ITAI 2372

Advantages of Using AI in the Manufacturing Industry

Artificial Intelligence (AI) is changing many industries around the world. One of the most important industries is manufacturing. AI in manufacturing is the use of machine learning (ML) and deep learning to improve manufacturing processes. These technologies help companies analyze large amounts of data and make better decisions.

Today, many factories use smart machines and Internet of Things (IoT) devices. These devices collect a lot of information every day, such as machine temperature, and production speed. AI can study this information very quickly and find patterns that people may not see. This helps companies solve problems faster and improve the production process. AI is also an important part of Industry 4.0, that focuses on automation, smart factories, and connected machines.

There are many ways AI is used in manufacturing. It can predict machine failures before they happen, check product quality with cameras, help robots work more accurately, reduce waste, and improve supply chain management. These applications help companies save time, reduce costs, increase productivity, and produce better-quality products.

The successful implementation of AI systems can bring many benefits to the manufacturing industry. First, AI can improve performance and decision-making by analyzing large amounts of data and providing useful information. This allows managers and company leaders to make faster and better decisions based on real data. Another important benefit is the improvement of human work. AI can also support employees by helping them work faster and more efficiently. When workers have access to smart tools and accurate information, they can solve problems more easily and improve their productivity over time. AI can also create safer workplaces.

Many manufacturing tasks are repetitive, dangerous, or require a high level of attention. By using automation, robots, and intelligent systems, companies can reduce human exposure to risky situations while also reducing mistakes caused by manual processes.

But implementing AI in manufacturing is not always simple. One important challenge is employee adaptation. Workers need training to understand how to use new technologies effectively. Without proper education, companies may not receive the benefits of their AI systems. Another challenge is that new technologies can sometimes make company systems more complicated instead of improving them.

AI in Amazon's warehouses

Amazon is one of the largest online retailers in the world and also one of the biggest cloud computing companies. The company operates in more than 20 countries and serves hundreds of millions of customers every year. Because of this enormous scale, Amazon manages a huge number of products, warehouses, and deliveries every day. To support its global operations, Amazon has invested heavily in artificial intelligence (AI), robotics, and digital technologies. The company also created Amazon Web Services (AWS) in 2006, a cloud computing platform that provides services such as data storage, computing power, and databases for organizations around the world.

Before using advanced AI systems, many warehouse activities required significant human effort. Employees had to search for products, organize inventory, move items, and check products manually. Traditional warehouse methods became less efficient. The company needed to solve several problems. Amazon uses robotics and AI-powered inventory management systems. One example is Sequoia, a robotic inventory system designed to improve how products are identified and stored inside warehouses. This technology helps Amazon identify and store inventory approximately 75% faster.

Another challenge was product quality. When customers receive products that do not match their expectations, they may return them. For example, around 22% of product returns happen because items look different from what customers expected. Processing these returns creates additional costs for transportation, inspection, and storage. Amazon developed an AI system called Project P.I. (Private Investigator) to detect product defects before items are shipped to customers. The system combines computer vision and generative AI to check products and identify problems such as incorrect color, size, or visible damage. Before Project P.I., employees had to manually inspect products using a visual checking process. This required significant time and human effort.

Some benefits of implementing AI are:

- AI has improved efficiency and speed. The Sequoia inventory system allows Amazon to identify and store products much faster, improving warehouse organization and reducing processing time.
- Amazon has reduced human effort and employee injuries by approximately 15%.
- AI has helped reduce processing time by around 25%.

AI provides many advantages, implementing these systems but it also creates challenges. One major challenge is the high cost of developing and maintaining advanced technologies. Robots, AI software, and computer systems require significant investment. Another challenge is employee adaptation. Workers need proper training to understand how to use new technologies and work together with AI systems. Without training, companies may not achieve the expected results.

AI Worker Assistance System

An AI Worker Assistance System designed to support employees in warehouses and manufacturing environments. The main goal of this system is not to replace workers, but to help them perform their tasks more efficiently, safely, and independently. This AI system would be available to workers at any time through devices such as tablets. It would work as a personal AI agent that can provide information, answer questions, and guide employees.

The system would combine different AI technologies, including Internet of Things (IoT), Retrieval-Augmented Generation (RAG), Computer Vision, and Machine Learning. By using IoT data from machines and equipment, the AI could understand the current situation inside the workplace. Through RAG technology, the system could access company manuals, safety instructions, technical documents, and training materials to provide accurate answers. Computer Vision could help the AI analyze images or identify problems with products or machines.

One direct application of this technology would be helping workers solve problems related to specific procedures. For example, if an employee does not know how to correctly package a product or how to solve a small technical issue with a machine, they could ask the AI assistant and receive step-by-step instructions immediately.

This system could significantly reduce the amount of time workers spend looking for information or waiting for assistance from supervisors. With an AI assistant, workers could solve many problems independently while managers could focus on other tasks. Another important benefit is employee training. Traditional training programs can require significant time and financial investment. AI would reduce the investment in courses. But implementing this system would also create some challenges. Companies would need to invest in the necessary technology and ensure that the AI provides accurate information.

In conclusion, AI in manufacturing is no longer just a technology of the future, it is already the present. As we can see in the Amazon case study, AI is not only used to manage inventory inside warehouses, but it is involved in many different steps before a product arrives at our homes. AI helps recommend products when customers visit the Amazon website, organize and manage inventory in warehouses, and detect damaged products using computer vision systems. The results of AI implementation in Amazon’s operations are impressive. These technologies have improved efficiency, reduced risks in warehouses, and helped reduce the number of damaged products sent to customers. Because of the successful implementation of AI in manufacturing, I decided to focus my proposal on supporting workers. Employee assistance does not directly control production, but has a major impact on overall performance. Providing workers with an AI assistant can improve training, reduce mistakes, and give employees access to important information whenever they need it.

Arm Ltd. (n.d.). What is in AI Manufacturing? Arm | the Architecture for the Digital World. https://www.arm.com/glossary/ai-in-manufacturing

How is AI Used in the Manufacturing Industry. (2026, March 10). MIT Sloan Executive Education. https://executive.mit.edu/How-is-AI-used-in-the-manufacturing-industry.html

Awati, R., & Yasar, K. (2025, April 23). What is Amazon? Definition and company history of Amazon.com. WhatIs. https://www.techtarget.com/whatis/definition/Amazon

Team, S. (2024, July 2). How Amazon is using AI to become the fastest supply chain in the world. Sifted. https://sifted.com/resources/how-amazon-is-using-ai-to-become-the-fastest-supply-chain-in-the-world/